

# **Operating Systems**

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## **Report:**

### **Compile:**

```
gcc -Wall -o prog prog.c
```

### **Run:**

```
./prog input.1 FCFS
```

```
./prog input.1 RR 5
```

```
./prog input.1 SJF
```

### **Screenshots:**

**SJF**

**RR:**

```

matthew@LAPTOP-9I9622SL:~/CS4420/instruction-cs4420/Homework 4$ ./prog input.1 RR 5
RR:
Time 0: PID 0 running
Time 1: PID 0 running
Time 2: PID 0 running
Time 3: PID 0 running
Time 4: PID 0 running
Time 5: PID 1 running
Time 6: PID 1 running
Time 7: PID 1 running
Time 8: PID 1 running
Time 9: PID 1 finished
Time 9: PID 2 running
Time 10: PID 2 finished
Time 10: PID 3 running
Time 11: PID 3 running
Time 12: PID 3 finished
Time 12: PID 0 running
Time 13: PID 0 running
Time 14: PID 0 running
Time 15: PID 0 running
Time 16: PID 0 running
Time 17: PID 0 running
Time 18: PID 0 running
Time 19: PID 0 finished

PID  Arrival  Start  End  Running  Waiting
0      0        0    19   12       7
1      2        5     9    4        3
2      3        9    10    1        6
3      4       10    12    2        6

Average Waiting Time: 5.5

```

## FCFS:

```

● matthew@LAPTOP-9I9622SL:~/CS4420/instruction-cs4420/Homework 4$ gcc -Wall -o prog prog.c
● matthew@LAPTOP-9I9622SL:~/CS4420/instruction-cs4420/Homework 4$ ./prog input.1 FCFS
FCFS:
Time 0: PID 0 running
Time 1: PID 0 running
Time 2: PID 0 running
Time 3: PID 0 running
Time 4: PID 0 running
Time 5: PID 0 running
Time 6: PID 0 running
Time 7: PID 0 running
Time 8: PID 0 running
Time 9: PID 0 running
Time 10: PID 0 running
Time 11: PID 0 running
Time 12: PID 0 finished
Time 12: PID 1 running
Time 13: PID 1 running
Time 14: PID 1 running
Time 15: PID 1 running
Time 16: PID 1 finished
Time 16: PID 2 running
Time 17: PID 2 finished
Time 17: PID 3 running
Time 18: PID 3 running
Time 19: PID 3 finished

PID  Arrival  Start  End  Running  Waiting
0      0        0    12   12       0
1      2       12    16    4      10
2      3       16    17    1      13
3      4       17    19    2      13

Average Waiting Time: 9

```